

Functional genetic report · longevity

Microbiome and Gut

Sérgio F · Report date: 02-10-2026

As the genes shape what feeds the gut bacteria, the intestinal barrier, the immune response to bacteria, bile and TMAO. Only genes (chip + local imputation + polygenic scores).

Contents

• Summary	3
• How to read this report	4
• Polygenic scores	5
• FUT2 (secretor) and lactase: what feeds bacteria	6
• Intestinal barrier, celiac disease and inflammatory bowel disease	7
• Immunity of the intestine and histamine (TLR4, TLR5, DAO, HNMT)	9
• Biliary acids and vesicle	10
• TMAO (FMO3)	11
• Fiber, transit, irritable intestine and diverticules	12
• Mechanisms of ageing	13
• Practical suggestions	14
• Limitations	15
• Methods and sources	16

Summary

31

SNPs analyzed

28

in the chip

2

imputed ($DR2 \geq 0.8$)

1

SNP not found in the data

11

Polygenic scores

The microbiome is not inherited, but the host genes choose part of the terrain: mucus sugars (FUT2), lactose that reaches the colon (LCT), the way the immune system recognizes bacteria (NOD2, ATG16L1, CARD9, TLR) and bile that bathes. Dysbiosis and intestinal barrier loss feed low-grade chronic inflammation (inflammaging), and bacterial TMAO binds the intestine to the arteries.

In summary: the genetic terrain of the intestine is very favourable. Celiac disease is practically excluded (without HLA-DQ2.5 or DQ8; p4 score) and inflammatory bowel disease has a very low polygenic risk (p2 and p9), without the variants NOD2 of Crohn. It is secretory (FUT2) and maintains lactase. The weaknesses are small: CARD9 in the risk alleles, slightly reduced histamine degradation (DAO, HNMT), ATG16L1 T300A heterozygous and a vesicle stone score in the 72nd percentile.

Favourable

- **Celiac disease practically excluded:** HLA-DQ2.5 (rs2187668 CC) and DQ8 (rs7454108 TT) absent; celiac score in the 4th percentile.
- **Inflammatory bowel disease** scores in the 2 and 9 percentiles; NOD2 R702W and G908R absent; ulcerative colitis p27.
- **FUT2 Secretor** (rs601338 GA) and persistent lactase (rs4988235 GA): mucus with glycans that feed bifidobacteria; tolerated dairy products.
- **Bilis:** ABCG8 D19H absent (rs11887534 GG); simple score of stones in the vesicle in the 12th percentile.
- **OCTN1 L503F absent** (rs1050152 CC): normal ergothione transport; HNF4A without the risk allele.

Unfavourable

- **CARD9 S12N TT and rs10781499 AA:** risk alleles of inflammatory bowel disease (antifungal immunity).
- **Histamine:** DAO Thr16Met and His645Asp heterozygous, HNMT Thr105Ile heterozygous: a slightly reduced degradation of food histamine.
- **ATG16L1 T300A heterozygous** (autophagy of Paneth cells) and IL23R without the protective allele.
- **Vesicle:** Rock score (876 variants) in the 72nd percentile, and UGT1A1*28 heterozygous.
- **Amphibian disease** in the 72nd percentile; FMO3 without variants of low activity (the bacterial TMA is all converted to TMAO). NOD2 1007fs not visible.

Interpretation: The genetic risk of autoimmune or inflammatory bowel disease is among the lowest possible. Genetics leaves the intestine in the hands of the diet: very varied fiber, fermented fiber, few emulsifiers and care with TMAO and vesicle during rapid loss of adiposity.

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 28 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 0 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

The scores of celiac disease and inflammatory bowel disease are of complete genome and come from the analysis with imputation. The simple scores (irritable intestine, frequency of the discards, stones in the vesicle) use only the leading variants of the GWAS Catalog and are less accurate. The frequency score of the discards has no judgment (more frequent is not necessarily better).

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
Celiac disease	PGS000040	122 de 228 (54%)	p4	Favourable
Inflammatory bowel disease (139 variants)	PGS004105	137 de 139 (99%)	p2	Favourable
Inflammatory bowel disease	PGS001288	156 de 193 (81%)	p9	Favourable
Crohn's disease	PGS001331	211 de 256 (82%)	p48	Typical / no judgement
Ulcerative colitis	PGS001306	141 de 176 (80%)	p27	Typical / no judgement
Diseases of the appendix	PGS001853	19 de 19 (100%)	p62	Typical / no judgement
Amphibian disease	PGS000996	308 de 368 (84%)	p72	Typical / no judgement
Irritable intestine (GWAS Catalog score, 37 loci)	C_SII	37 de 37 (100%)	p66	Typical / no judgement
Dejection frequency (GWAS Catalog score, 10 loci)	C_FEZES	10 de 10 (100%)	p96	Typical / no judgement
Vesicle stones (876 variants)	PGS001256	772 de 876 (88%)	p72	Typical / no judgement
Vesicle stones (GWAS Catalog score, 35 loci)	C_VESICULA	33 de 35 (94%)	p12	Favourable

FUT2 (secretor) and lactase: what feeds bacteria

FUT2 puts H antigen (ABO sugars) in the mucus and secretions: secretors offer bacteria, especially bifidobacteria, mucus glycans as food. The persistence of lactase decides whether lactose is digested in the small intestine or fermented in the colon.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
FUT2 W154X (secretor)	rs601338	GA	chip	A	1	0.45	FUT2 W154X AA = non-secretor (without ABO antigens in the mucus; less bifidobacteria, more Crohn's disease; resistant to norovirus). 1 copy = secretor. (strong evidence)	Typical / no judgement
FUT2	rs602662	GA	chip	—	—	—	FUT2 (G258S, in connection with W154X): serum vitamin B12. (moderate evidence)	Typical / no judgement
LCT/MCM6 -13910C>T	rs4988235	GA	chip	G	1	0.46	Lactase persistence: allele A (T-13910) maintains lactase in adults; GG : non-persistent (lactose intolerance). (strong evidence)	Typical / no judgement
LCT/MCM6 -22018G>A	rs182549	CT	chip	C	1	0.46	According to the marker of lactase persistence (in connection with -13910). Persistence is associated with less bifidobacteria when there is consumption of dairy products. (moderate evidence)	Typical / no judgement

Favourable

- FUT2 rs601338 GA: secretor (1 allele non-secretor)
- LCT rs4988235 GA and rs182549 CT: persistent lactase (tolerant)

Unfavourable

- None

Interpretation: Heterozygous secretor: intermediate mucous profile, the most common. With persistent lactase, lactose does not reach the colon; bifidobacteria depend more on fiber (inulin, GOS) and fermented ones.

Intestinal barrier, celiac disease and inflammatory bowel disease

Celiac disease (HLA-DQ2.5/DQ8), Crohn's disease and ulcerative colitis (NOD2, ATG16L1, IRGM, IL23R, CARD9, PTGER4, HNF4A). These genes also describe the quality of the barrier and the response to bacteria in healthy people.

Polygenic scores of this theme: Celiac disease (PGS000040): the 4th percentile **Favourable**; Inflammatory bowel disease (139 variants) (PGS004105): the 2nd percentile **Favourable**; Inflammatory bowel disease (snpgnet) (PGS001288): the 9th percentile **Favourable**; Crohn's disease (PGS001331): the 48th percentile **Typical / no judgement**; Ulcerative colitis (PGS001306): the 27th percentile **Typical / no judgement**; Attachment diseases (PGS001853): the 62nd percentile **Typical / no judgement**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
HLA-DQA1 (DQ2.5)	rs2187668	CC	chip	T	0	0.01	HLA-DQ2.5 marker, the main genetic factor of celiac disease. (strong evidence)	Favourable
HLA-DQB1 (DQ8)	rs7454108	TT	chip	C	0	0.11	HLA-DQ8 marker (according to the haplotype of celiac disease). (strong evidence)	Favourable
NOD2 R702W	rs2066844	CC	chip	T	0	0.06	NOD2 R702W: bacterium peptidoglycan sensor; loss of function → Crohn ileal disease, less defensins. (strong evidence)	Favourable
NOD2 G908R	rs2066845	GG	chip	C	0	0.01	NOD2 G908R: Crohn's disease. (strong evidence)	Favourable
NOD2 1007fs	rs2066847		SNP not found in the data	—	—	—	NOD2 1007fsinsC: the strongest Crohn variant (insertion). (strong evidence)	SNP not found in the data
ATG16L1 T300A	rs2241880	AG	chip	G	1	0.55	ATG16L1 T300A: autophagy of less efficient Paneth cells (less defensins, more Crohn). (moderate evidence)	Typical / no judgement
IRGM	rs13361189	TT	chip	—	—	—	IRGM (autophagy of intracellular bacteria): Crohn disease. (moderate evidence)	Typical / no judgement
IL23R R381Q	rs11209026	GG	chip	A	0	0.06	IL23R Gln381: protector (~2–3x minus Crohn, colitis and psoriasis). (strong evidence)	Typical / no judgement
CARD9 S12N	rs4077515	TT	chip	T	2	0.41	CARD9 (antifungal immunity, Dectin-1): risk allele of IBD. (moderate evidence)	Unfavourable
CARD9	rs10781499	AA	chip	A	2	0.41	CARD9: risk allele of inflammatory bowel disease. (moderate evidence)	Unfavourable
SLC22A4 (OCTN1) L503F	rs1050152	CC	chip	T	0	0.40	Ergothione OCTN1 transporter: allele 503F associated with Crohn and less ergothione in tissues. (weak evidence)	Favourable
PTGER4	rs4613763	TT	chip	C	0	0.14	EP4 receiver of prostaglandin E2 (epithelial repair): IBD. (moderate evidence)	Typical / no judgement
HNF4A	rs6017342	AA	chip	C	0	0.52	HNF4A (differentiation of the epithelium and tight junctions): ulcerative colitis. (moderate evidence)	Favourable
MYO9B	rs2305764	GG	chip	—	—	—	MYO9B: intestinal permeability (inconsistent association with celiac/IBD). (weak evidence)	Typical / no judgement

Favourable

- HLA-DQ2.5 and DQ8 absent (celiac score p4)
- NOD2 R702W and G908R absent
- IBD p2 and p9 scores; ulcerative colitis p27; HNF4A AA
- OCTN1 rs1050152 CC

Unfavourable

- CARD9 rs4077515 TT and rs10781499 AA
- ATG16L1 T300A AG (typical)
- IL23R R381Q absent (as in most cases)

Interpretation: Without DQ2.5 or DQ8, celiac disease is almost impossible (negative predictive value >99%); gluten has no genetic contraindications. The coverage of the celiac score is 54% (HLA region), but the two direct HLA markers confirm. Crohn (p48 score) is typical; the global IBD is very low. NOD2 1007fs (insertion) is not in the data.

Immunity of the intestine and histamine (TLR4, TLR5, DAO, HNMT)

TLR4 detects bacterial LPS; TLR5 flagellin. DAO (AOC1) degrades food histamine in the intestinal lumen; HNMT degrades it inside the cells.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
TLR4 D299G	rs4986790	AA	chip	G	0	0.05	TLR4 Gly299: weaker response to bacterial LPS. (weak evidence)	Typical / no judgement
TLR5 R392X	rs5744168	GG	chip	A	0	0.06	TLR5 stop (not detect flagellin): less inflammation, but composition of the different microbiome. (moderate evidence)	Typical / no judgement
AOC1 (DAO) Thr16Met	rs10156191	CT	chip	T	1	0.27	Diamine oxidase (degrades food histamine in the intestine): Met16 reduces the activity. (moderate evidence)	Typical / no judgement
AOC1 (DAO) His645Asp	rs1049793	CG	chip	G	1	0.27	DAO Asp645: less activity. (moderate evidence)	Typical / no judgement
HNMT Thr105Ile	rs11558538	CT	chip	T	1	0.11	Histamine N-methyltransferase Ile105: minor activity. (weak evidence)	Unfavourable

Favourable

- TLR4 D299G absent (normal response to LPS)

Unfavourable

- DAO Thr16Met CT and His645Asp CG (less activity)
- HNMT Thr105Ile CT (unfrequent)

Interpretation: Three slower alleles of histamine degradation, all heterozygotes. No symptoms have no consequence; if there is a reaction to foods rich in histamine, this is the probable mechanism. DAO needs copper, vitamin B6 and vitamin C.

Biliary acids and vesicle

Biliary acids digest the fats and are hormonal signals (FXR, TGR5) that bacteria transform into secondary bile acids. ABCG5/8 export cholesterol to bile; BSEP exports bile salts; CYP7A1 converts cholesterol into bile acids.

Polygenic scores of this theme: Vesicle stones (876 variants) (PGS001256): the 72nd percentile
Typical / no judgement; Vesicle stones (GWAS Catalog score, 35 loci) (C_VESICULA): the 12th percentile
Favourable

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
ABCG8 D19H	rs11887534	GG	chip	C	0	0.07	ABCG8 His19: more cholesterol in the bile → stones in the vesicle (~2x per allele); less sterol absorption. (strong evidence)	Favourable
ABCB11 (BSEP) V444A	rs2287622	AG	chip	—	—	—	Biliary salt pump BSEP: cholestasis and stones. (moderate evidence)	Typical / no judgement
CYP7A1 -204A>C	rs3808607	GG	chip	—	—	—	Biliary acid synthesis limiting enzyme from cholesterol: response of LDL to diet. (weak evidence)	Typical / no judgement
UGT1A1 *80 (*28 marker)	rs887829	CT	chip	T	1	0.30	Gilbert: higher bilirubin → more pigmentary stones (small effect in heterozygous). (moderate evidence)	Typical / no judgement
SULT2A1	rs212100	CC	imputation DR2=1.00	—	—	—	Biliary acid sulphate: stones in the vesicle. (weak evidence)	Typical / no judgement

Favourable

- ABCG8 D19H absent
- Simple score of stones (35 loci) p12

Unfavourable

- Rock score (876 variants) p72
- UGT1A1*28 heterozygous (pigmentary stones, small effect)

Interpretation: The two stone scores disagree (p12 and p72): the one of 876 variants is more complete, so the risk is typical, slightly pulled up. Retalutida and rapid loss of adiposity increase the risk of stones in any person.

TMAO (FMO3)

The bacteria transform choline, carnitine and betaine into TMA; the liver converts TMA into TMAO through FMO3. TMAO is associated with atherosclerosis and thrombosis.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
FMO3 E158K	rs2266782	GG	chip	A	0	0.37	FMO3 Lys158: ~ minor → less conversion of TMA (from the bacteria, from choline and carnitine) into TMAO. (moderate evidence)	Typical / no judgement
FMO3 E308G	rs2266780	AA	chip	G	0	0.17	FMO3 Gly308: lower activity (especially in combination with Lys158). (moderate evidence)	Typical / no judgement
SLC44A1/ choline	rs3199966	TT	imputation DR2=0.98	—	—	—	Choline transport (TMA bacterial substrate). (weak evidence)	Typical / no judgement

Favourable

- None.

Unfavourable

- None.

Interpretation: FMO3 Glu158/Glu158 and Glu308/Glu308: normal activity (without variants that lower the conversion). Thus, TMAO depends mainly on the diet and TMA-producing bacteria. With 9p21 risk and a score of venous thrombosis p86 (cardiovascular report), it is worth keeping the TMAO low.

Fiber, transit, irritable intestine and diverticules

The fiber feeds the bacteria that produce butyrate (colon fuel and barrier). There are no well-established fiber response genes; transit scores, irritable intestines and diverticules are the best genetic proxy.

Polygenic scores of this theme: Funicular disease (PGS000996): the 72nd percentile Typical / no judgement; Irritable intestine (GWAS Catalog score, 37 loci) (C_SII): the 66th percentile Typical / no judgement; Dejection frequency (GWAS Catalog score, 10 loci) (C_FEZES): the 96th percentile Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
Favourable					Unfavourable			
• None.					• None.			

Interpretation: Frequency of discharges in the 96th percentile: genetic trend for faster transit (less constipation). Irritable p66 and diverticular disease p72: typical, high side. Fiber protects against diverticular disease, so more fiber is always better.

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
Dysbiosis	CARD9 risk; Normal FMO3 (TMAO depends on the diet)	FUT2 secretory; persistent lactase Normal NOD2	Favourable genetic terrain.
Intestinal Barreira dysfunction	ATG16L1 T300A het; Absence IL23R	HLA-DQ2.5/DQ8 absent; IBD p2/p9 Normal NOD2; HNF4A and OCTN1 without risk	Very favourable.
Chronic Inflammation / Inflammation	CARD9 TT/AA; histamine (DAO, HNMT)	Celiac and very low IBD; normal TLR4	Favourable.
Autophagy / Deregulated Macroautophagy	ATG16L1 T300A heterozygous	IRGM without defined risk	Small local weakness (paneth cells).
Vascular Aging	FMO3 of normal activity (plus TMAO with diet rich in carnitine/choline)	—	Manage by diet.
Metabolic Homeostasis Decline	P72 stone score; UGT1A1*28	ABCG8 D19H absent; single score p12	Typical.
Intercellular Communication	Reduced histamine degradation (3 alleles)	—	Only relevant with symptoms.
Neuroinflammation	—	Genetically robust intestinal barrier (gut-brain axis)	Favourable.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
Very varied fiber	More fiber, always: 30+ different plants per week; legumes, oats/brews (beta-glucans), onion, garlic, garlic, asparagus (inulin), resistant starch (potato/cooled rice), psyllus. <i>suggestion (not applied)</i>	P72 diverticular disease; IBD low; FUT2 secretor
Fermented	1-2 portions/day of natural yogurt, kefir, sauerkraut or other fermented vegetables (increased diversity and lower inflammatory markers). Dairy products are tolerated (persistent lactase). <i>suggestion (not applied)</i>	Persistent LCT; secretory
Emulsifiers	Limit ultraprocessed with carboxymethylcellulose, polysorbato-80 and carrageenan, which refine the mucus and favor bacteria that cross it. <i>suggestion (not applied)</i>	CARD9 risk; ATG16L1 het
TMAO	Red meat with moderation; avoid supplementation of L-carnitine and choline/ phosphatidylcholine in high dose; extra virgin olive oil (inhibes the bacterial production of TMA) and fiber. <i>suggestion (not applied)</i>	FMO3 of normal activity; 9p21 and thrombosis (cardiac report)
Vesicle during adiposity loss	With the retathrutide, regular meals with some fat (the vesicle needs to contract), fiber and without prolonged fasting with zero fat. If the loss is rapid, discuss with the doctor the preventive deoxycholic acid. <i>suggestion (not applied)</i>	P72 stone score; UGT1A1*28
Histamine (only with symptoms)	If there is reaction, prefer fresh food and avoid leftovers, cured cheeses, sausages and fermented long for a few days; maintain copper, B6 and vitamin C of the food (DAO cofactors). <i>suggestion (not applied)</i>	DAO x2 and HNMT heterozygotes
Gluten	Without genetic restriction: with DQ2.5 and DQ8 absent, celiac disease is practically excluded (whole grains count for fiber). <i>suggestion (not applied)</i>	HLA-DQ2.5/DQ8
Protecting the microbiome	Avoid antibiotics and proton pump inhibitors without clear indication; sleep and training regularly (both increase diversity). <i>suggestion (not applied)</i>	General coordination
Polyphenols	Cocoa, red fruits, pomegranate, green tea and olive oil feed beneficial bacteria (Akkermansia, bifidobacteria). Pausing curcumin, luteolin, apigenin and quercetin while the ALT is high is a suggestion (not applied; the real stack has not changed). <i>suggestion (not applied)</i>	V2 plane rules
Coordination with medication	LDN, which already takes, has benefit data in the intestinal barrier (Crohn's disease); the retathractide slows gastric emptying, so fiber and water avoid constipation. D3 10 000 IU and omega-3 3.2-4 g/day as in the plane. No telomerase activators. <i>suggestion (not applied)</i>	Current medicine

Limitations

- The microbiome itself is not genetic: this report describes the host's terrain, not the bacteria present (for this it would be necessary a fecal analysis).
- The insertion NOD2 1007fs, the number of copies of AMY1 (salivary amylase) and the complete HLA alleles are not read by the chip; HLA is inferred by markers.
- The celiac score covers 54% of the variants (HLA region); the simple scores of irritable intestine, transit and stones use few leading variants.
- The percentiles compare with the reference population (n=633).
- Only genes: does not use blood or feces analysis.

Methods and sources

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (Beagle 5.5, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs). Frequency of alleles calculated locally in the reference population ($n=633$); the effect allele of each SNP was confirmed by the expected frequency.
- Alleles of effect cured from literature or GWAS Catalog (local copy of 13-09-2026), with the verified sense.
- Scores: PGS000040 (celiac), PGS004105 and PGS001288 (IBD), PGS001331 (Crohn), PGS001306 (ulcerous colitis), PGS001853 (appendix), PGS000996 (diverticles), PGS001256 (vesicle stones) and simple GWAS Catalog scores for irritable intestine (37 loci), frequency of the dejections (10 loci) and stones (35 loci). Percentages vs. reference population ($n=633$) with exactly the same variants.
- Main references: Monsuur 2008 (rs2187668 and DQ2.5); Hugot 2001 and Ogura 2001 (NOD2, Nature); Hampe 2007 (ATG16L1, Nat Genet); Duer 2006 (IL23R, Science); McGovern 2010 and Rausch 2011 (FUT2); Bonder 2016 and Goodrich 2016 (host genetics and microbiome); Buchmann 2007 (ABCG8 D19H); Wang 2011 and Tang 2013 (TMAO, Nature/NEJM); Maintz 2007 (DAO); Chassaing 2015 (emulsifiers, Nature); Wastyk 2021 (fermented, Cell).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).